Protocol/Statistical Analysis Plan: I8B-MC-ITRR

Pharmacokinetics and Glucodynamics of LY900014 Compared to Insulin Lispro Following Single Dose Administration in Elderly and Younger Adults With Type 1 Diabetes Mellitus

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Pharmacokinetics and Glucodynamics of LY900014 Compared to Insulin Lispro Following Single Dose Administration in Elderly and Younger Adults with Type 1 Diabetes Mellitus

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LY900014

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1. Protocol Synopsis

Title of Study:

Pharmacokinetics and Glucodynamics of LY900014 Compared to Insulin Lispro Following Single Dose Administration in Elderly and Younger Adults with Type 1 Diabetes Mellitus

Rationale:

LY900014 is an ultra-rapid-acting insulin lispro formulation with increased initial absorption compared to commercially available insulin lispro formulation (Humalog®; Eli Lilly and Company). LY900014 aims at mimicking the physiological prandial insulin secretion pattern better than the currently available treatments, which may more effectively control postprandial glucose excursions and allow increased flexibility in the time of dosing relative to a meal.

The aim of this study is to compare the insulin lispro pharmacokinetic (PK) and glucodynamic (GD) profiles of LY900014 with those of insulin lispro (Humalog) during a euglycaemic glucose clamp in elderly and younger adults with type 1 diabetes mellitus (T1DM).

Objective(s)/Endpoints:

Objectives	Endpoints
Primary • To evaluate the PK of insulin lispro following administration of a single 15-U SC dose of LY900014 compared to insulin lispro (Humalog) within each T1DM population (elderly and younger adults).	• Early 50% t_{max} and $AUC_{(0\text{-}30min)}$
Secondary • To compare the GD during a euglycaemic clamp of LY900014 and insulin lispro (Humalog) following administration of a single 15-U SC dose in elderly and younger adults with T1DM. • To evaluate the safety and tolerability of the novel LY900014 formulation.	• Early 50% TR_{max}, $Gtot_{(0\text{-}30min)}$, $Gtot_{(0\text{-}1hour)}$, and T_{onset} • AEs

Abbreviations: AE = adverse event; $AUC_{(0\text{-}30min)}$ = area under the concentration versus time curve from time zero to 30 minutes; early 50% t_{max} = time to early half-maximal plasma concentration; early 50% TR_{max} = time to early half-maximal glucose infusion rate; GD = glucodynamics; $Gtot_{(0\text{-}30min)}$ = total amount of glucose infused over 30 minutes; $Gtot_{(0\text{-}1hour)}$ = total amount of glucose infused over 1 hour; PK = pharmacokinetics; SC = subcutaneous; T1DM = type 1 diabetes mellitus; T_{onset} = time to onset of insulin action.

Summary of Study Design:

Study I8B-MC-ITRR is a randomized, subject- and investigator-blind, 2-treatment, 2-period, crossover study in elderly and younger adults with T1DM.

Treatment Arms and Duration:
LY900014 (test): Single 15-U dose
Insulin lispro (Humalog) (reference): Single 15-U dose

Number of Subjects: Up to 42 younger adult (aged 18 to 45 years inclusive) and 42 elderly (aged ≥65 years) subjects may be enrolled to ensure that at least 34 subjects in each age group complete the study.

Statistical Analysis: The primary statistical analyses for PK will be conducted on subjects who receive at least 1 dose of study drug and have measurable insulin lispro concentrations. The primary statistical analysis for GD will be conducted on those subjects who complete at least 1 clamp procedure. Supportive analyses will be done on the key parameters for the subjects who complete all treatment periods with evaluable data. Safety analyses will be conducted for all enrolled subjects who received at least 1 dose of study drug, whether or not they completed all protocol requirements.

Unless otherwise noted, all tests of treatment effects will be conducted at a 2-sided alpha level of 0.05, and confidence intervals (CIs) will be calculated at 95%, 2-sided.

Safety: All investigational and protocol procedure adverse events (AEs) will be listed, and if the frequency of events allows, safety data will be summarized using descriptive methodology. Safety parameters that will be assessed include safety laboratory parameters, vital signs, and electrocardiogram parameters as well as injection-site reactions. The parameters will be listed and summarized using standard descriptive statistics.

Pharmacokinetics: Subjects who receive at least 1 dose of study drug and have measurable insulin lispro concentrations will be included in the PK analysis dataset. PK parameter estimates for insulin lispro will be calculated using standard noncompartmental methods of analysis.

Log-transformed maximum observed concentration (C_{max}), clearance, and area under the plasma concentration-time curve estimates for insulin lispro will be evaluated to estimate least-squares geometric means, ratios of geometric means between LY900014 and Humalog (reference), and their corresponding 95% CIs for each age group using the statistical model that includes treatment, sequence, and period as fixed effects and subject within sequence as a random effect. The same model without log transformation will be used for time to early half-maximal plasma concentration (early 50% t_{max}), time to late half-maximal plasma concentration (late 50% t_{max}), and time to maximum concentration (t_{max}) to estimate least-squares means, treatment differences in least-squares means, and the corresponding 95% CIs for each age group. The treatment ratios and the corresponding 95% CIs for these time parameters will be calculated using Fieller's theorem.

As treprostinil plasma concentrations after administration of LY900014 are expected to be very low, a descriptive statistical analysis may be conducted if sufficient data are available for this analysis. The primary analysis of the treprostinil concentrations will be focused on assessing t_{max} and C_{max}.

Glucodynamics: GD assessments will be determined from the glucose clamp procedure, where the glucose infusion rate (GIR) over time will be used as a measure of insulin effect. GD analyses will be conducted on those subjects who complete at least 1 clamp procedure.

A locally weighted scatterplot smoothing (LOESS) function will be applied to all individual GIR versus time profiles in each treatment group and/or period. The fitted data for each subject will be used to calculate the GD

parameters. The values of these GD parameters will be summarized by treatment and/or period through descriptive statistics. Mean LOESS fits of GIR versus time profiles will be generated.

The GD statistical model will be similar to the model used for the analysis of the PK parameters. The variables of maximum GIR (R_{max}), total amount of glucose infused (Gtot), total amount of glucose infused over 30 minutes ($Gtot_{[0\text{-}30min]}$), total amount of glucose infused over 1 hour ($Gtot_{[0\text{-}1h]}$), and total amount of glucose infused over 10 hours ($Gtot_{[0\text{-}10h]}$) will be log-transformed prior to analysis. The time variables including time to onset of insulin action (T_{onset}), time to R_{max} (TR_{max}), time to half-maximal GIR before TR_{max} (early 50% TR_{max}), and time to half-maximal GIR after TR_{max} (late 50% TR_{max}) will be analyzed using the same model without log transformation to estimate least-squares means, treatment differences in least-squares means, and the corresponding 95% CIs. The treatment ratios and the corresponding 95% CIs for these time parameters will be calculated using Fieller's theorem.

Immunogenicity:

The immunogenicity data will be listed. The relationship between treatment-emergent antidrug antibodies and AEs will be evaluated. The relationship between the presence of antibodies and the PK parameters and GD response to insulin lispro may also be evaluated.

2. Schedule of Activities

Study Schedule: Protocol I8B-MC-ITRR

Procedures	Screening	Euglycaemic Clamp Procedure Periods 1 and 2		FU/ED[a]	Instructions/Comments
	Up to Day -28	Day -1	Day 1		
Informed consent[b]	X				
Subject admission to CRU		X			
Standardized dinner		X			All subjects will receive short-acting insulin either via an injection or via a bolus dose from the insulin pump before the start of a standardized dinner and 6 hours prior to dosing.
Fasting	X	X		X	Subjects are expected to fast for at least 8 hours before screening and FU, and at least 8 hours before each dose until after the glucose clamp procedure is completed. (Up to 20 g of carbohydrates will be allowed to prevent hypoglycaemia.)
Height	X				
Weight	X		Predose for Period 1 and Period 2	X	
Waist and hip circumference[d]		X			
Alcohol breath test	X	X		X	
Run-in and stabilization of glucose			X		To start approximately 1-6 hours prior to study drug dosing.
Medical assessment and physical examination	X	X	Before discharge from CRU for each period	X	Physical examination and medical history at screening. Thereafter, medical review and targeted examination, as appropriate.
Vital signs: body temperature, blood pressure, and pulse rate	X	X	Predose, 0.5 and 2 hours after dose and at the end of the clamp procedure	X	Vital signs can be collected after the subject is supine for at least 5 minutes. Body temperature at screening only. Predose vital signs may be collected up to 2 hours before the start of the clamp procedure.
Randomization		X			Only at Period 1

Procedures	Screening	Euglycaemic Clamp Procedure Periods 1 and 2		FU/ED[a]	Instructions/Comments
	Up to Day -28	Day -1	Day 1		
12-Lead ECG	X		Predose and at the end of the clamp procedure	X	Predose ECGs can be collected up to 2 hours before the start of the clamp procedure. Subjects must be supine for at least 10 minutes before ECG collection and remain supine but awake during ECG collection.
Clinical laboratory tests	X		Predose for Period 1 only	X	Fasting laboratory test for screening, Period 1, and FU. To monitor patient safety, additional tests may be performed at the discretion of the investigator, as needed throughout the study.
Pregnancy test/urine drug screen[c]	X	X		X	Pregnancy test for all female subjects. Urine drug screen for all subjects at CRU admission.
Study drug administration			X		Time = 0 minutes
Injection-site local tolerability assessments			0, 60, and 240 minutes postdose and at the end of clamp procedure		Time 0: assessments of injection-site reaction will occur immediately following the injection.
VAS assessment of injection site pain			0, 20, and 60 minutes postdose		Time 0: assessments of injection-site pain will occur immediately following the injection.
Insulin lispro PK sampling			0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 90, 120, 150,180, 240, 300, 360, 420, 480, 540, and 600 minutes		Sampling times are relative to study drug administration time (0 minutes).
Treprostinil PK sampling			15, 30, 60, and 120 minutes		Sampling times are relative to study drug administration time (0 minutes).
Blood glucose sampling for euglycaemic clamp			Blood glucose monitoring every minute starting from the run-in and throughout the duration of the clamp until 10 hours after dosing		
Pharmacogenetic sample			Predose for Period 1 only		Refer to sample collection instructions provided by the sponsor.
Immunogenicity sample			Predose for Period 1 and Period 2	X	
Discharge from CRU			X		After completion of all study procedures and medical assessment on Day 1. Investigator's discretion to discharge on Day 2.

Abbreviations: CRU = clinical research unit; ECG = electrocardiogram; ED = early discontinuation; FU = follow-up; ICF = informed consent form; PK = pharmacokinetic; VAS = visual analog scale.

[a] Procedures should be performed at least 14 days after the last dose of study drug.

[b] Screening shall be completed within 28 days of dose of study drug. The ICF will be signed at least 1 day prior to screening.

[c] A FSH test will be performed for women who are menopausal

[d] The average of triplicate measurements.

NOTE: The site should schedule activities as appropriate. In cases where several study procedures are scheduled at the same time, the order of priority will be as follows: PK samples including blood sampling for blood glucose and laboratory samples per protocol nominal times. ECGs and vital sign measurement should be scheduled before but as close as possible to the PK sampling times. VAS and injection-site assessments can be done after PK sampling If the investigator decides based on clinical judgment not to dose a subject on a given day (eg, because of low blood glucose), the subject's visit may be rescheduled; any procedures performed in that period may be repeated.

3. Introduction

3.1. Study Rationale

The use of LY900014 in elderly subjects (≥65 years of age) is anticipated. Study I8B-MC-ITRR (ITRR) is being conducted in accordance with regulatory guidance to conduct clinical trials of medicines that are likely to have significant use in the elderly.

LY900014 is an ultra-rapid-acting insulin lispro formulation, which has shown an increased initial absorption compared to commercially available insulin lispro formulation (Humalog®, Eli Lilly and Company). LY900014 aims at mimicking the physiological prandial insulin secretion pattern, which may more effectively control postprandial glucose excursions and allow increased flexibility of the time of dosing relative to a meal.

A population of elderly subjects is selected as changes in the body composition due to aging (such as increased body fat, reduced lean body mass, decreased total body water) may alter the pharmacokinetic (PK) profile of insulin lispro in LY900014.

Minimal data are available for the evaluation of the PK and glucodynamics (GD) of insulin lispro after LY900014 administration in subjects ≥65 years of age. Therefore, the aim of this study is to compare the PK and GD profiles of insulin lispro from LY900014 with those of insulin lispro (Humalog) during a euglycaemic glucose clamp in elderly subjects and younger adults with type 1 diabetes mellitus (T1DM) and also in a population of younger adults to support prescribing information and gain further data in this population.

3.2. Background

The insulin analog insulin lispro (Humalog) has been shown to be absorbed more quickly than regular human insulin (Humalog package insert, 2015). In healthy volunteers given subcutaneous (SC) doses of insulin lispro ranging from 0.1 to 0.4 units/kg, peak serum levels were seen 30 to 90 minutes after dosing (Humalog package insert, 2015). However, the general consensus is that rapid-acting insulin, administered either through pumps or syringes/pen injectors, is still not rapid enough to match carbohydrate absorption profiles, which limits efficacy and dosing flexibility. An ultra-rapid-acting prandial insulin would shift the PK/GD of insulin analogs so that they have an even faster onset to better match carbohydrate absorption and also may allow important flexibility in the time of dosing relative to meals.

LY900014 represents a new formulation that contains insulin lispro, treprostinil, citrate, and other excipients. This formulation involves the novel use of a small amount of treprostinil (Remodulin®) as an excipient to enhance the absorption of insulin lispro by local vasodilatation rather than as an active pharmaceutical ingredient to elicit a systemic effect. Treprostinil is a prostacyclin analog, administered either through inhalation (Tyvaso®), as an intravenous (IV) infusion, or as a continuous SC administration for the treatment of symptomatic pulmonary arterial hypertension and has been approved in the United States (US) since 2002 and in Germany since 2007 (Remodulin package insert, 2014). Each of the other excipients (such as sodium citrate and magnesium chloride) in the LY900014 formulation is listed in the US Food and Drug Administration's (FDA's) Generally Recognized as Safe Food Additives database and

in the FDA's Inactive Ingredients in Approved Drugs database. Furthermore, the excipient concentration in LY900014 is within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

Safety and tolerability of LY900014 has been demonstrated in healthy subjects in 2 previous clinical studies, CCI . Both were Phase 1, randomized, subject-blind studies in which the PK and GD of LY900014 and Humalog insulin lispro were evaluated during a euglycaemic glucose clamp following SC administration. The total insulin lispro exposure and GD effect were similar for LY900014 and Humalog insulin lispro; however, LY900014 demonstrated a faster and earlier insulin lispro absorption and faster elimination compared to Humalog insulin lispro.

Preliminary data from 3 ongoing Phase 1b studies CCI CCI showed LY900014 was well tolerated in subjects with T1DM using either multiple daily injections (MDI) or insulin pump treatment. There were no serious adverse events (SAEs) related to study treatment or discontinuations from the studies because of a drug-related adverse event (AE). Small numbers of treatment-emergent adverse events were reported, and there were no notable increases in these events in relation to any of the LY900014 formulations compared to those in relation to Humalog.

More information can be found in the Investigator's Brochure (IB) of LY900014.

3.3. Benefit/Risk Assessment

Study I8B-MC-ITRR will not offer any direct benefits to the subjects participating in the study. The data from previous studies CCI and preliminary data from ongoing studies CCI have shown that LY900014 was well tolerated and the adverse drug reactions are in keeping with those reported for Humalog.

Potential risks associated with LY900014, derived from the known risks of insulin lispro (Humalog)- the active pharmaceutical ingredient in LY900014, are hypoglycaemia, hypersensitivity reactions (localized allergy and/or systemic allergy), undesirable effects at the injection site (injection-site reactions and lipodystrophy), and peripheral edema (Humalog package insert, 2015).

Notably, across all doses in the Lilly clinical studies that have evaluated treprostinil as a local vasodilator with or without insulin lispro, there was no clinically significant increase in those AEs associated with systemic absorption of treprostinil, as described in the Remodulin package insert (2014) (that is, headache, diarrhea, nausea, jaw pain, vasodilatation, rash, edema, anorexia, vomiting, asthenia, abdominal pain, and hypotension). The exposures of treprostinil in LY900014 for participants in this study are expected to be undetectable compared to those observed in the dose ranges previously explored with SC bolus administration of treprostinil and are expected to be substantially lower than those observed in the treatment of pulmonary arterial hypertension.

No additional potential risks of LY900014 or treprostinil alone were identified in preclinical safety pharmacology and toxicity studies or clinical pharmacology studies. No known potential risks are associated with the use of small amounts of treprostinil (Remodulin) in the LY900014 formulation.

Additionally, local and systemic toxicity profiles of Humalog and Remodulin do not suggest the potential for additive or synergistic toxicity.

Following administration of the study insulins, subjects will receive IV glucose infusion at a variable rate to maintain euglycaemia up to 10 hours after insulin lispro administration. The aim of the clamp procedure is to maintain blood glucose within the normal glycaemic range. In addition, the clamp is performed while the subject is inpatient and under the investigator's supervision. These considerations should minimize the risk of hypoglycaemia in subjects participating in Study ITRR.

More information about the known and expected benefits, risks, SAEs, and reasonably anticipated AEs of LY900014 is to be found in the IB.

More detailed information about the known and expected benefits and risks of Humalog insulin lispro may be found in the Humalog package insert (2015).

4. Objectives and Endpoints

Table ITRR.1 shows the objectives and endpoints of the study.

Table ITRR.1. Objectives and Endpoints

Objectives	Endpoints
Primary • To evaluate the PK of insulin lispro following administration of a single 15-U SC dose of LY900014 compared to insulin lispro (Humalog) within each T1DM population (elderly and younger adults).	• Early 50% t_{max} and $AUC_{(0\text{-}30min)}$
Secondary • To compare the GD during a euglycaemic clamp of LY900014 and insulin lispro (Humalog) following administration of a single 15-U SC dose in elderly and younger adults with T1DM. • To evaluate the safety and tolerability of the novel LY900014 formulation.	• Early 50% TR_{max}, $Gtot_{(0\text{-}30min)}$, $Gtot_{(0\text{-}1hour)}$, and T_{onset} • AEs
Exploratory • To evaluate whether the difference between LY900014 and insulin lispro in PK and PD parameters is similar between elderly and younger adults.	• Early 50% t_{max} and $Gtot_{(0\text{-}30min)}$

Abbreviations: AE = adverse event; $AUC_{(0\text{-}30min)}$ = area under the concentration versus time curve from time zero to 30 minutes; early 50% t_{max} = time to early half-maximal plasma concentration; early 50% TR_{max} = time to early half-maximal glucose infusion rate; GD = glucodynamics; $Gtot_{(0\text{-}30min)}$ = total amount of glucose infused over 30 minutes; $Gtot_{(0\text{-}1h)}$ = total amount of glucose infused over 1 hour; PD = pharmacodynamics; PK = pharmacokinetics; SC = subcutaneous; T1DM = type 1 diabetes mellitus; T_{onset} = time to onset of insulin action.

5. Study Design

5.1. Overall Design

This is a Phase 1, randomized, subject- and investigator-blind, 2-treatment, 2-period, crossover study that compares the PK and GD of insulin lispro from LY900014 and insulin lispro (Humalog) following administration of a single 15-U SC dose in elderly and younger adults with T1DM who are either on multiple daily insulin injections or on CSII via a sponsor-approved insulin pump. Study ITRR will be conducted at 1 or more sites.

Figure ITRR.1 illustrates the study design.

Subjects will be randomized to 1 of 2 treatment sequences according to the actual randomization table provided to the site. Each subject will participate in a screening visit, 2 clinical research unit (CRU) inpatient dosing visits, and a follow-up visit. Before each dose, a run-in period of 1 to 6 hours to stabilize blood glucose will occur. Dosing visits will be separated by a wash-out period of 3 to 15 days, during which subjects will resume their normal insulin treatment with the exemption of the insulin switch 48 to 72 hours prior to dosing as mentioned below. At each visit, subjects will receive either LY900014 or insulin lispro and undergo a euglycaemic clamp procedure where the time-concentration and time-action profiles of the study treatment will be evaluated simultaneously for up to 10 hours using an automated clamp device. Briefly, the aim of the euglycaemic clamp is to maintain euglycaemia after the administration of a dose of insulin by means of variable glucose infusion. The variable glucose infusion maintains or "clamps" glucose at a constant euglycaemic target; therefore, the glucose infusion rate (GIR) reflects the GD effect of the insulin.

Prior to each dosing visit, subjects with T1DM on multiple daily insulin injections have to discontinue their basal insulin, according to the following guidance:

- For subjects using insulin degludec or insulin glargine U300, the last injection of insulin degludec should occur no later than 72 hours prior to dosing.
- For subjects using insulin detemir or glargine, the last injection of insulin detemir or glargine should occur no later than 48 hours prior to dosing.
- For subjects using neutral protamine Hagedorn (NPH) insulin or other intermediate-acting insulin, the last injection of NPH insulin or other intermediate-acting insulin should occur no later than 24 hours prior to dosing.
- Subjects requiring any infusion via CSII of insulin lispro (Humalog) will switch to insulin glulisine (Apidra®) at least 8 hours prior to dosing. Basal infusion rate via CSII should not occur later than 3 hours prior to dosing.
- Any injection or bolus infusion via CSII of more than 6 U of short-acting insulin should not occur between 11 and 6 hours prior to dosing. Any bolus injection or bolus infusion via CSII should occur no later than 6 hours prior to dosing.

All subjects will be instructed to perform at least 4-point daily self-monitoring plasma glucose (SMPG), as well as continue their short-acting insulin during lead-in. The SMPG and prandial insulin doses should be recorded in the patient's diary, which will serve as a source document.

5.1.1. Inpatient Dosing Visits

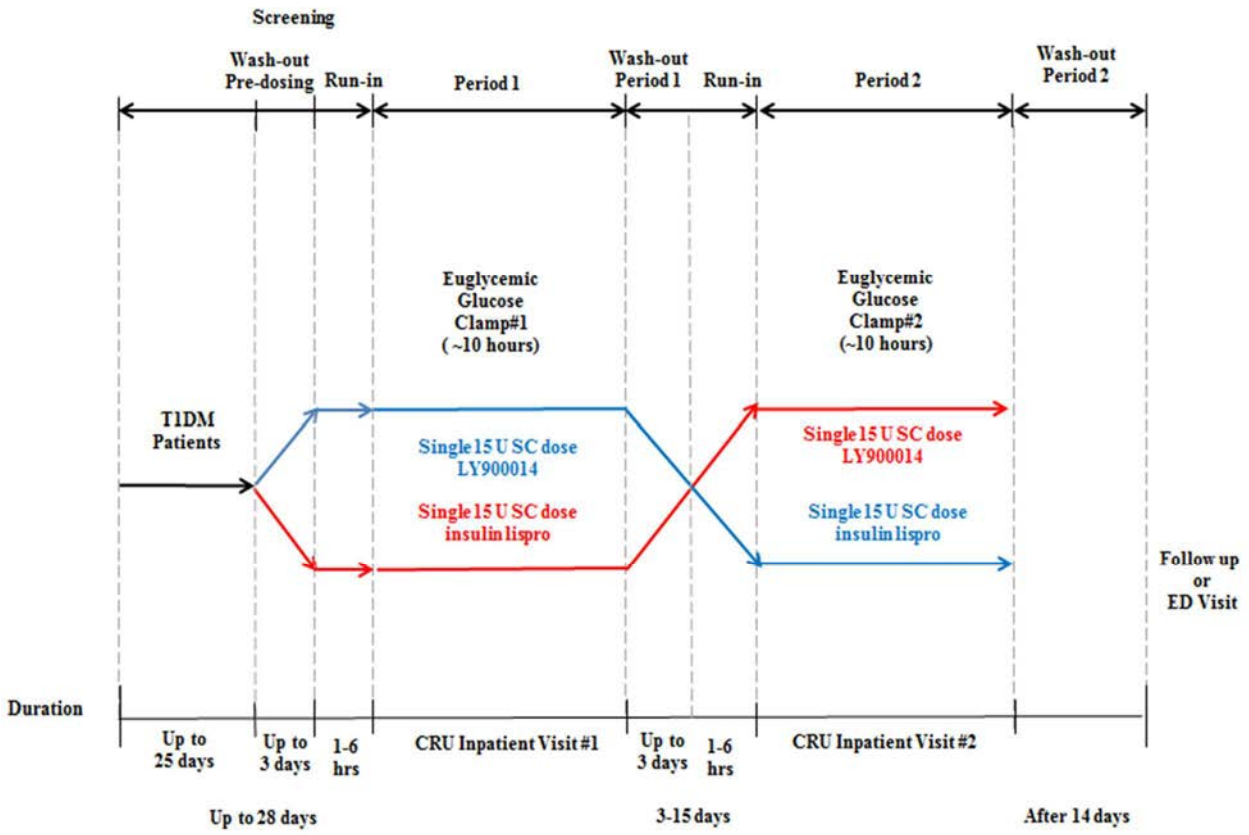
The patient will check into the CRU on Day -1 of each period in the early evening (at approximately 1700 hours). Subjects on multiple daily insulin injections would have discontinued their basal insulin but subjects using pump therapy (CSII) should arrive at the CRU with their basal rate running. All subjects will receive short-acting insulin either via an injection or via a bolus dose from the insulin pump before the start of a standardized dinner within the CRU. This prandial dose should be administered no later than 6 hours prior to the planned study drug dosing the next day. After dinner (at approximately 2200 hours), the subject is required to fast until the completion of the clamp procedure the following day. Consumption of food or beverages other than water later than approximately 2200 hours on the evening before dosing is not allowed, with the exception of minor intake of rapidly absorbable carbohydrates (not more than 20 g) if necessary to prevent hypoglycaemia (if carbohydrates have been ingested, subjects' blood glucose measurement will have to confirm that they are not hypoglycaemic). If hypoglycaemia (blood glucose level ≤3.5 mmol/L [≤ 63 mg/dL] or plasma glucose ≤3.9 mmol/L [≤70 mg/dL]) or more than 20 g of carbohydrates are needed to prevent hypoglycaemia less than 24 hours prior to dosing, the dosing visit can be rescheduled 1 to 7 days later. Each of the dosing visits can only be rescheduled once.

Prior to study treatment dosing, the patient will be connected to the clamp device for continuous glucose monitoring and the start of the baseline run-in period. Variable IV infusion of either glucose (20% dextrose solution) or insulin (insulin glulisine [Apidra]) will be started to reach a target blood glucose level. Once the target blood glucose level is attained and remains stable, with the predose activities as specified in the Schedule of Activities completed, the patient will receive a dose and undergo the clamp procedure proper. More information on the clamp methodology can be found in Section 9.6.

5.1.2. Outpatient Procedure

After completing each period, subjects will resume their previous insulin regimen. The investigator will provide instructions for transition back to a normal insulin regimen. Subjects will return for the second dosing period within a 3- to 15-day interval, during which they will have to perform at least 4-point daily SMPG. The basal insulin will have to be discontinued as described previously.

After completing all study treatment dosing and clamp procedures, subjects will resume their former insulin regimen and SMPG routine. A CRU visit for follow-up or early discontinuation should occur at least 14 days after the last dose of study drug.



Abbreviations: CRU = clinical research unit; ED = early discontinuation; SC = subcutaneous; T1DM = type 1 diabetes mellitus.

Figure ITRR.1. Illustration of study design for Protocol I8B-MC-ITRR.

5.2. Number of Participants

Up to 42 younger adult and 42 elderly subjects may be enrolled to ensure that at least 34 subjects in each age group complete the study.

If subjects drop out of the study before completion of both clamp periods, replacement subjects may be enrolled; the replacement subjects will assume the same treatment sequence as the subjects who dropped out and will complete that treatment sequence in its entirety.

5.3. End of Study Definition

End of study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient in Period 2.

5.4. Scientific Rationale for Study Design

The use of a crossover design allows each patient to serve as his or her own control, thereby reducing variability. The study is subject- and investigator-blind to minimize potential bias.

A euglycaemic clamp technique, the gold standard methodology for assessing insulin action, will be used in this study to provide data on the GD activity of each study insulin lispro formulation (see Section 9.6 for a detailed description of the clamp methodology).

A population of elderly subjects is selected as changes in the body composition due to aging (such as increased body fat, reduced lean body mass, decreased total body water) may alter the PK profile of insulin lispro in LY900014.

Based on the PK properties of treprostinil ($t_{1/2}$ = approximately 1 hour) and Humalog ($t_{1/2}$ = 0.79 hours), the duration of the washout period between clamp visits (a minimum of 3 days) and the duration between the last dose of study drug and the follow-up visit (at least 14 days) are considered appropriate.

5.5. Justification for Dose

Based on previous studies of both insulin lispro (Humalog) and LY900014, the 15-U dose is within the clinical dose range and should provide measurable PK and GD profiles for both study insulins. The safety, PK, and pharmacology of LY900014 at similar doses and with similar formulation composition have been assessed in 2 clinical studies in healthy subjects CCI and in ongoing studies in patients with T1DM using MDI CCI or insulin pump treatment CCI and type 2 diabetes mellitus using MDI CCI In patients with T1DM, the weight-based total daily insulin dose for a large majority of patients is 0.4 to 0.8 U/kg/day of which approximately 50% is delivered as the basal insulin dose and the other 50% is divided among preprandial insulin doses (Herbst and Hirsch 2002; Heller et al. 2009). Therefore, a reasonably high estimate for total daily preprandial insulin dose in T1DM is 0.4 U/kg/day. It is estimated that the lowest body weight for either population would be around 56 kg. Thus, the 15-U dose of LY900014 within this study would delivery approximately a 0.26-U/kg dose and would not exceed this high estimate for the total daily preprandial insulin dose.

A 15-U dose of LY900014 contains CCI of treprostinil in a single SC dose, which is within the range evaluated as safe and tolerated in all previous studies CCI . The PK of treprostinil in LY900014 following SC administration of 15 U of LY900014 was assessed previously in CCI , treprostinil exposure was not detectable for the 15-U dose of LY900014 CCI Based on the treprostinil exposures observed in CCI , treprostinil blood concentrations in this study are expected to be CCI

6. Study Population

Eligibility of subjects for study enrollment will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests, and electrocardiogram (ECG).

The nature of any conditions present at the time of the physical examination and any preexisting conditions will be documented.

Screening may occur up to 28 days prior to enrollment. Subjects who are not enrolled within 28 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

6.1. Inclusion Criteria

Subjects are eligible for inclusion in the study only if they meet all of the following criteria at screening and/or enrollment:

[1] are male or female subjects with T1DM for at least 1 year. A diagnosis of T1DM is based on medical history with a fasting C-peptide ≤0.30 nmol/L.

[1a] Male subjects:

agree to use an effective method of contraception for the duration of the study and for 1 month following the last dose of the investigational product.

[1b] Female subjects:

women of childbearing potential may participate and include those who test negative for pregnancy before initiation of treatment based on a serum pregnancy test and agree to use 1 highly effective method of contraception or a combination of 2 effective methods of contraception during the study and for 1 month following the last dose of the investigational product.

women of nonchildbearing potential may participate in the study without using adequate contraceptive methods, and include those who are:

- infertile due to surgical sterilization (hysterectomy, bilateral oophorectomy, or tubal ligation), congenital anomaly such as mullerian agenesis;
- or postmenopausal, defined as women aged <52 years and being amenorrhoeic for more than 1 year with a serum follicle-stimulating hormone (FSH) level compatible with postmenopausal status or aged ≥52 years and being amenorrhoeic for less than 1 year and with a serum FSH level compatible with postmenopausal status or aged ≥52 years being amenorrhoeic for more than 1 year

[2] are aged 18 to 45 years (both inclusive) (younger adult group) or at least 65 years (elderly group) at the time of screening.

[3] have a body mass index (BMI) of 18.5 to 30.0 kg/m^2, inclusive, at screening.

[4] have clinical laboratory test results within normal reference range for the population or investigative site, or results with acceptable deviations that are judged to be not clinically significant by the investigator.

[5] have venous access sufficient to allow for glucose infusion and blood sampling procedures as per protocol.

[6] are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures.

[7] are able and willing to give signed informed consent approved by Lilly and the ethical review board (ERB) governing the site.

[8] have a glycated hemoglobin (HbA_1c) <9.0% at screening.

[9] have had no episodes of severe hypoglycaemia in the last 6 months (see Section 9.4.7.1).

6.2. Exclusion Criteria

Subjects will be excluded from study enrollment if they meet any of the following criteria at screening and/or enrollment:

[10] are investigative site personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child, or sibling.

[11] are Lilly employees or employees of the investigational site/CRU.

[12] are currently enrolled in a clinical trial involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study.

[13] have participated within the last 30 days in a clinical trial involving an investigational product. If the previous investigational product has a long half-life, 3 months or 5 half-lives (whichever is longer) should have passed.

[14] have previously completed or withdrawn from this study.

[15] have known allergies to treprostinil (Remodulin), insulin lispro, insulin glulisine (Apidra), related compounds, or any components of the formulation, or a history of significant atopy.

[16] have an abnormality in the 12-lead ECG that, in the opinion of the investigator, increases the risks associated with participating in the study. Elderly patients with clinically significant bradycardia will be excluded.

[17] have a supine blood pressure at screening outside the range of 90 to 140mmHg for systolic or 50 to 95 mmHg for diastolic (1 repeat is allowed) as determined by the investigator, or results with unacceptable deviations that are judged by the investigator to be clinically significant for the population, or have a heart rate outside the range of 50 to 90 beats/minute.

[18] have a history or current cardiovascular, respiratory, hepatic, renal, gastrointestinal, endocrine (apart from T1DM), hematological, or neurological disorders capable of significantly altering the absorption, metabolism, or elimination of drugs; of constituting a risk when taking the investigational product; or of interfering with the interpretation of data.

[19] have known or ongoing psychiatric disorders.

[20] regularly use known drugs of abuse and/or show positive findings on urinary drug screening.

[21] show evidence of an acute infection with fever or infectious disease at the time of study entry.

[22] show evidence of human immunodeficiency virus (HIV) infection and/or positive human HIV antibodies.

[23] show evidence of hepatitis C and/or positive hepatitis C antibody (the presence of hepatitis C antibodies in the setting of normal liver function tests and a negative hepatitis C polymerase chain reaction are not an exclusion).

[24] show evidence of hepatitis B and/or positive hepatitis B surface antigen (the presence of antibodies to the hepatitis B surface antigen is not an exclusion).

[25] are women who are pregnant or lactating.

[26] have significant lipohypertrophy in the target abdominal injection area as judged by the investigator.

[27] have, except for current regimen of insulin therapy and concomitant medication(s) (for example, antihypertensive medication, lipid-lowering agent, thyroid hormone replacement medication, hormonal contraception, hormonal replacement therapy), regular use of or intended use of any over-the-counter or prescription medications or nutritional supplements that treat hyperglycaemia or insulin resistance or that promote weight loss within 14 days before dosing.

[28] are currently a smoker, used tobacco products on a regular basis in the 6 months before screening, or are intending to smoke during the study period.

[29] are receiving chronic (lasting longer than 14 consecutive days) systemic or inhaled glucocorticoid therapy (excluding topical, intra-articular, and intraocular preparations), or have received such therapy within the 4 weeks before screening.

[30] have donated blood of more than 450 mL or more in the last 3 months or provided any blood donation within the last month before screening.

[31] have a significant history of alcoholism or drug abuse as judged by the investigator or consuming more than 24 g of alcohol per day for men, or more than 12 g of alcohol per day for women (1 unit of alcohol is defined as 10 mL [8 g] of pure alcohol).

[32] are unwilling to comply with the dietary requirements/restrictions during the study: (i) comply with the fasting requirements of the study, (ii) consume only the meals/snacks provided during the inpatient visits.

[33] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study.

[34] have a history of renal impairment (exclusion only if glomerular filtration rate [estimated GFR] <60 mL/minute/1.73 m^2 [GFR is estimated according to the Chronic Kidney Disease Epidemiology Collaboration creatinine equation], or have a serum creatinine level ≥126 µmol/L [male] or ≥111 µmol/L [female]).

[35] have a history of deep vein thrombosis of the leg or repeated episodes of deep leg vein thrombosis in first-degree relatives (parents, siblings, or children) as judged by the investigator.

[36] have proliferative retinopathy or maculopathy and/or severe neuropathy; in particular, autonomic neuropathy as judged by the investigator based on a recent (<1.5 years) ophthalmologic examination.

[37] any significant changes in insulin regimen and/or unstable blood glucose control within the past 3 months prior to screening as assessed by the investigator.

[38] require daily insulin treatment >1.5 U/kg/body weight.

6.3. Lifestyle and/or Dietary Requirements

Throughout the study, subjects may undergo medical assessments and review of compliance with requirements before continuing in the study.

6.3.1. Meals and Dietary Restrictions

In both treatment periods, subjects will fast for at least 8 hours prior to dosing until after the glucose clamp procedure is completed, after which subjects will receive a meal. Water can be consumed freely during this time. If the clamp procedure is completed in less than 10 hours, then the meal should be delayed until the last PK sample has been taken, unless the investigator deems it necessary to administer the meal for safety reasons.

While resident in the CRU, subjects may not consume any food or caloric drinks other than that provided by the CRU. When not resident in the CRU, subjects will be encouraged to follow their normal diets.

6.3.2. Caffeine, Alcohol, and Tobacco

Subjects should refrain from caffeine-containing food/beverages (for example, cola, chocolate drinks, tea, coffee, energy drinks containing methylxanthine [caffeine, theophylline, or theobromine]) for at least 12 hours before each dose and throughout the duration of each CRU visit.

No alcohol will be allowed at least 24 hours before each CRU admission (Day –1) and throughout the duration of each CRU visit (see Additional Exclusion Criterion [31]). Between dosing visits, daily alcohol should not exceed 3 units for males and 1.5 units for females (a unit is defined in Exclusion Criterion [31], Section 6.2).

Smoking (cigars, cigarettes, or pipes) and the use of smokeless tobacco will not be permitted during the study (see Exclusion Criterion [28]).

6.3.3. Activity

Subjects are encouraged to maintain their regular exercise habits; however, subjects engaging in strenuous physical exercise (for example, weightlifting, football, cycling/swimming/running at a level higher than normal) within 48 hours prior to (planned) dosing will be excluded from this study, as judged by the investigator to interfere with study results. After dosing, subjects should remain recumbent or sitting in the CRU until the end of the glucose clamp. Movement will be restricted to retain the integrity of connections to the infusion(s) and the study procedures.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study may be rescreened 1 time. Any further rescreening must be done in consultation with the sponsor. Individuals who still do not meet participation criteria after rescreening (screen failure) may not be rescreened.